Rob Basnet

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Education

Gettysburg College Gettysburg, PA
Bachelor of Science in Computer Science; Minors in Mathematics and Data Science *Expected May 2027*

- **Relevant Coursework:** Machine Learning, Probability & Statistics, Linear Algebra, Calculus III, Data Structures & Algorithms, Distributed Systems, Operating Systems

Technical Skills

Languages: Python, Java, C++, C, SQL, JavaScript/TypeScript
Quant & Data: pandas, NumPy, scikit-learn, statsmodels, TA-Lib, backtesting, walk-forward validation, probability calibration, time series analysis
Systems: FastAPI, PostgreSQL, Redis, Docker, WebSockets, React, Git, REST APIs, CI/CD
Financial Data: Alpaca, Polygon, Finnhub, Tiingo, Yahoo Finance, SEC EDGAR, market microstructure concepts

Research Experience

Independent Quantitative Research Gettysburg, PA
Machine Learning vs. Linear Factor Combination in US Large-Cap Equities *2026 – Present*

- Designing a pre-registered out-of-sample study testing whether machine learning (gradient boosting, random forests) combines momentum, value, and quality signals into a stronger long–short portfolio than an equal-weight linear composite, net of costs.
- Established the baseline using a self-built backtesting engine: the equal-weight composite earns ≈0 out-of-sample Sharpe after an 8 bps cost model, under walk-forward validation with purge/embargo to prevent leakage.
- Building the ML pipeline and comparison harness scoring Sharpe, deflated Sharpe, drawdown, and turnover; deliverable is a 6–8 page paper with a public replication repo.

Projects

Multi-Factor Equity Backtesting Framework | *Python, pandas, NumPy*

- Built a point-in-time, survivorship-bias-free backtesting engine for momentum, value, and quality factors by constructing historical index membership and lagged fundamentals from SEC EDGAR and price APIs.
- Implemented decile long–short portfolio construction with an 8 bps transaction cost model, producing turnover-adjusted Sharpe, deflated Sharpe, and drawdown analytics benchmarked against SPY.
- Validated every strategy with walk-forward out-of-sample folds and purge/embargo windows, surfacing that in-sample factor performance largely disappears out-of-sample after costs.

AI Stock Analytics & Signal Platform | *Python, FastAPI, React, PostgreSQL, Redis*

- Reduced API call volume by 70% and eliminated rate-limit incidents by integrating four market data providers (Alpaca, Polygon, Finnhub, Tiingo) behind a unified failover layer with Redis caching, serving real-time quotes and portfolio analytics through FastAPI and Redis-cached WebSocket streams.
- Engineered a multi-factor signal engine combining EMA crossovers, market regime detection, sector rotation strength, and earnings-proximity filters, with a context-aware explanation for every signal change.

Premier League Match Prediction & Betting Model | *Python, scikit-learn, React*

- Improved log-loss versus uncalibrated baselines by applying Isotonic Regression calibration to five Random Forest classifiers trained on 5+ seasons of match data (W/D/L, expected goals, Over 2.5/3.5, BTTS markets).
- Produced tradeable edge estimates by blending calibrated model probabilities with market-implied odds priors, surfaced through a React dashboard backed by a FastAPI inference service.

Leadership Experience

Resident Assistant Lead Aug 2025 – Present
Gettysburg College *Gettysburg, PA*

- Increased floor participation by 40% by mentoring 38+ residents through conflict resolution and organizing 5+ community events per semester.

Calculus Peer Learning Assistant Sep 2024 – Present
Gettysburg College *Gettysburg, PA*

- Moved students from below-average to above-average exam performance by tutoring 10+ students weekly on limits, derivatives, and integrals using structured problem-solving frameworks.

Certifications & Additional

Virtual Experience: JPMorgan Chase Quantitative Research (Forage); Bank of America Global Markets (Forage)
Activities: ACM Member — hackathons and algorithmic coding challenges in Python, Java, and C++ (Sep 2023 – Present)
Self-Study: Hull, *Options, Futures, and Other Derivatives*; ongoing reading of SSRN factor-investing literature